



GIS How-Tos

How to use RESTful API with GIS Ethernet products

This document will provide a brief overview of the RESTful API conventions commonly used to communicate with the GIS Ethernet platform of products

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Overview

This document gives a brief introduction to the RESTful API conventions commonly used to communicate with the GIS Ethernet platform of products. It covers:

- Conventions used
- How you might use it and useful tools
- Glossary of terms

How to use RESTful API with GIS Ethernet products

- i** Some knowledge of HTTP is required to understand the RESTful interface
- i** Please see [Glossary](#) for common RESTful API references and terms used throughout this document
- i** Please see www.odata.org for further information on REST and OData references

RESTful Interface

A RESTful interface uses HTTP requests to allow a client to communicate with a server.

Examples include:

- Receiving data - monitoring the state of the printhead and print jobs
- Sending data - uploading waveforms and print data
- Executing commands - turning the printhead on or off, or aborting a print

HTTP API Basics

To make an HTTP request, the client must provide the URI for a server resource - beginning with `http://<server>:5000/api`. The request will also contain a method (e.g. GET, PUT, POST, DELETE, OPTIONS) and may include additional information

- i** To begin using the RESTful interface, a client needs an application capable of making HTTP requests - see [Useful Tools](#)
- i** Please see [REST overview on Wikipedia](#) for further information

API Documentation

The GIS RESTful API is accessed via a built-in UI browser using the Swagger open standard.

The Swagger file provides the full list of methods available through the GIS RESTful API - [GIS Swagger documentation](#)

- i** For a link to the Swagger open standard, see [Useful Tools](#)

Resource Structure

The RESTful API is presented as a tree structure, similar to the directories of a PC file system. An example diagram is shown below:

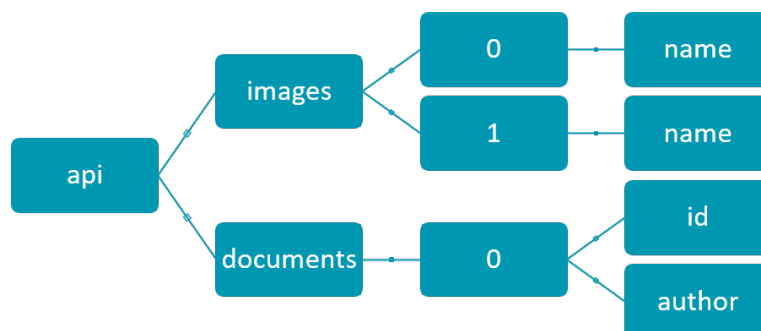


Figure 2-1 A simplified example of a RESTful API structure

The resources use the types that are available in JSON. Objects are used to represent complex types, such as printheads. Arrays are used to represent collections, such as the collection of waveforms. Other primitives (numbers, strings, Booleans) are used to represent the data stored in complex types, such as the waveform's name.

Using the example from Figure 1, a client can construct a URI that can be used to perform an operation on any part of the tree structure. For example:

- A GET operation on `http://HOST/api/documents/0` would serialize the document resource at index 0. This would return a JSON object which would look something like this:

```
{
  "id": "0",
  "author": "Alice"
}
```

- A GET operation on `http://HOST/api/images/1` would serialize the image resource at index 1. This would return a JSON object which would look something like this:

```
{
  "name": "my image 1",
}
```

- Instead of getting a single resource in the list, the client can “go up” a level and do a GET on the whole list. A GET operation on `http://HOST/api/images` would return the whole list:

```
[
  {
    "name": "my image 0",
  },
  {
    "name": "my image 1"
  }
]
```

- Going up another level and performing a GET on the `http://HOST/api` resource will return the whole API:

```
{
  "images": [
    {
      "name": "my image 0",
    },
    {
      "name": "my image 1"
    }
  ],
  "documents": [
    {
      "id": "0",
      "author": "Alice"
    }
  ]
}
```

Long Polling

Many resources provide a LongPollUri resource. This looks like a normal URI, but has a query parameter called “longpoll”, which is a string value.

- Performing a GET operation using the long poll URI will start an HTTP request. However, this request will not return until the state of the resource has changed from the state identified by the longpoll token
- Each resource has its own definition of what counts as a 'change'. Some resources have 'state' parameters; a change in this state will trigger a long poll change

Examples

Common Operations

The most common operations are:

- Create a new job
 - POST /api/jobs – job parameters are provided as multipart form data
- Add additional parameters to a job
 - POST /api/jobs/<id>/additionaldata – add more parameters as multipart form data
- Control and Monitor the job
 - GET /api/jobs/<id>
 - GET /api/jobs/<id>?longpolluri=... to wait for a change
 - GET /api/jobs/longPollUriAndValues- to get the current collection of jobs and a URI to wait for changes to the collection
 - POST /api/jobs/<id>/cancel
- Control and monitor a JobStore
 - GET /api/jobstores
 - GET /api/ jobstores/<id>
 - POST /api/ jobstores /<id>/cancel – cancel running jobs and leave the store stopped
 - POST /api/ jobstores /<id>/stop
 - POST /api/ jobstores /<id>/start

Low-level Operations

The low-level operations that may sometimes useful are:

- Upload some bitmap images
 - POST /api/bitmaps
 - POST /api/bitmaps/form
- Create a SeparatedImage
 - POST /api/separatedimages
 - POST /api/separatedimages/form

- either uploading the images as part of those requests, or referencing existing ones

- Create a PrintOperation
 - POST /api/printoperations
- Control and Monitor the PrintOperation
 - GET /api/printoperations/<id>
 - GET /api/printoperations/<id>?longpolluri=... to wait for a change
 - POST /api/printoperations/<id>/cancel

- Control and monitor a PrintQueue (currently the default is the only one)
 - GET /api/printqueues
 - GET /api/printqueues/default
 - POST /api/printqueues/default/stop
 - POST /api/printqueues/default/start
- Clean up
 - DELETE /api/bitmaps/<id>
 - DELETE /api/separatedimages/<id>

Useful Tools

The following tools may be of use when working with the GIS RESTful API for GIS Atlas products

Curl

Curl for Windows is a command-line utility that allows clients to make customised HTTP requests, either as a single request or as part of a script. The [Curl for Windows](#) program may be freely downloaded .

Firefox

Firefox allows the user to configure the maximum number of open connections to a single server . Most browsers typically limit this to 10-20. This may not be enough for a JavaScript application that uses AJAX request to long poll every print operation.

Insomnia

This is a REST client that gives more control over HTTP requests than a web browser . Different methods (GET , PUT , POST , etc.) can be used, the HTTP headers can be customised, and the request and response can be optionally parsed as JSON.

Swagger

This is a tool that can use [Swagger](#) documentation to generate a software API library, which makes HTTP requests and parses the responses. Many different programming languages are supported. The code that is generated is generally quite extensive, as it includes functionality that is not always used throughout GIS products. It is not necessary to use a generated API; examples supplied by GIS use a basic Python HTTP library for simplicity.



Raw Swagger is not very easy to read, but a Swagger editor can generate documentation in several formats

Wireshark

Wireshark allows the user to inspect network traffic on a local machine. This can be useful for debugging applications on the PC , as every request and response to and from the source can be inspected.

Glossary

Term	Definition
API	Application Programming Interface - enables different systems to interact with each other programmatically
BLOB	A Binary Large Object (BLOB) is a collection of binary data stored as a single entity in a database
Body	Additional data - this may be plain text or binary data depending on the request
CamelCase	CamelCase is a naming convention in which the first letter of each word in a compound word is capitalised - it is most often used in programming languages and website names
Curl	Command Line utility often used to interact with REST API endpoints. Used in documentation for request code samples. It is usually the default format used to display requests in API documentation
DELETE	The HTTP command method for deleting resources with a REST API
Endpoint	The URI that goes after the base URL and points towards the requested API functionality - indicating how you access the resource
GET	The HTTP command method for retrieving resources from a REST API
Header	Text only, key value pair
HTTP	Hypertext Transfer Protocol - how website and APIs communicate over the internet
HTTPS	How website and APIs communicate securely over the internet
JSON	JavaScript Object Notation. A data format commonly used for API request parameters and response body
LongPollUri	A URI that can be used as a substitute long polling id for long poll requests
Method	Command that specifies how you want to interact with the resource, such as GET, POST, etc.
POST	The HTTP command method for creating resources with a REST API
PUT	The HTTP command method for updating resources with a REST API
Query Parameter	Delimited key value pair at the end of a URI
Request	The way information is returned from an API - the client provides a resource URL with the proper authorisation to an API server, which then returns a response with the information requested
Resource	The information returned by an API
Response	The information returned by an API after a request is made - responses are usually in JSON or XML format
REST API	Also <i>RESTful API</i> ; representational State Transfer - uses web protocols (HTTP) to make requests and provide responses that are language agnostic - i.e. both the requests and responses are made over HTTP
Swagger	A mechanism for describing a RESTful API, enabling it to be displayed in an interactive way in a web browser
URI	Unique Resource Identifier - URIs are identifiers, and that can mean name (URN), location (URL), or both

Contact / Support - information

We have a dedicated team of in-house experts who pride themselves on providing you with world-class support. If you require further information on GIS products, then please contact us by any of the following methods:

GIS Support Method	Contact
Log a support request / enquiry	GISsupport@globalinkjetsystems.com
Learn more about GIS	www.globalinkjetsystems.com

Revision History

Rev.	Date	Author	Comment
1.0	25/06/18	NC	Conversion and Expansion into Flare
1.1	12/01/21	NC	Amended Support Contact details